

Simulating Error-Mitigated Quantum Game Theory on NISQ Devices: A Battle-of-the-Sexes Replication and a Prisoner's-Dilemma Extension with Guided Circuit Mapping

Muhammad Wajih Hyder
Dept. of Computer Science
FAST-NUCES
Karachi, Pakistan
wajihhyder22@gmail.com

Aazar Arnold
Dept. of Computer Science
FAST-NUCES
Karachi, Pakistan
aazarnold@email.com

Faizan Ali Abdulali
Dept. of Computer Science
FAST-NUCES
Karachi, Pakistan
fali.abdulali@email.com

Abstract—Quantum game theory extends classical game theory by giving players quantum strategies and shared entanglement, allowing them to reach equilibrium outcomes that no classical mixed strategy can achieve, however, running these games on real quantum hardware is not straightforward. Noisy Intermediate-Scale Quantum (NISQ) devices introduce gate errors, readout errors, and crosstalk that push measured payoffs away from their theoretical values. This paper replicates the quantum Battle of the Sexes from Diaz Agreda et al. in simulation, implements their Guided Circuit Mapping (GCM) error-mitigation strategy, and extends the whole pipeline to the Prisoner's Dilemma, which the original paper did not cover. The simulation uses a calibrated NISQ noise model in Qiskit-Aer with 31 entanglement values and 20 repetitions per strategy. We confirm the quantum equilibrium payoff of 2.5 in the Battle of the Sexes, a 108 percent improvement over the classical mixed-strategy equilibrium of 1.2. GCM lowers the root-mean-square error by 35 to 45 percent across all four strategies, and a paired t-test across the entanglement sweep finds every reduction highly significant. In the Prisoner's Dilemma, the quantum miracle move reaches the cooperative payoff of 3, a 200 percent gain over the classical defect equilibrium of 1, and GCM reduces error by 12 to 46 percent there as well. Since this work runs entirely in simulation and not on a real processor, the absolute errors are lower than those reported for real hardware, but the relative benefit of GCM holds in both cases.

Index Terms—quantum game theory, NISQ, Battle of the Sexes, Prisoner's Dilemma, error mitigation, Guided Circuit Mapping, Eisert-Wilkens-Lewenstein

I. INTRODUCTION

Quantum game theory is the study of how classical two-player games change when the players are given quantum strategies and a shared entangled state. Classical game theory, in its original form, studies how rational agents should act when each player's outcome depends on what the others choose. What quantum mechanics adds to this picture is entanglement, which allows the players to reach equilibria that no classical mixing of strategies can produce. This combination of quantum mechanics and strategic decision making is broadly known as quantum game theory.

This work was completed as a course project at FAST-NUCES, Karachi.

The Eisert-Wilkens-Lewenstein (EWL) protocol, introduced in 1999 [3], was the first systematic way to embed a two-player game in quantum mechanics. Eisert, Wilkens, and Lewenstein showed that for the Prisoner's Dilemma there is a quantum strategy that escapes the classical defect-defect trap when both players use it. The same protocol has since been applied to the Battle of the Sexes and other two-player games. In each case entanglement improves coordination and raises the expected payoff above the classical equilibrium.

Running these protocols on real hardware is not straightforward, however. Current quantum processors sit in the NISQ regime, which means they have enough qubits to run small circuits but not the error correction needed to keep the noise down. Gate errors, decoherence, and crosstalk between neighbouring qubits build up quickly, so the measured payoffs drift away from the theoretical curve. Hence, without some form of error mitigation it is hard to confirm any quantum advantage on a real device.

Diaz Agreda et al. addressed this problem directly in their paper "Bridging Theory and Practice in Quantum Game Theory" [4]. They ran the quantum Battle of the Sexes on IBM's `ibm_sherbrooke` processor and proposed Guided Circuit Mapping (GCM) as an error-mitigation strategy. GCM profiles a set of qubit-pair slots on the device, routes each circuit to the lowest-error slot, and keeps the pairs well separated so that crosstalk stays low. Their results showed that GCM brought the experimental payoffs noticeably closer to the analytical curve. However, they ran only five repetitions and left a formal statistical test of the improvement as future work.

In this paper, we build on that work in three ways. First, we replicate the Battle of the Sexes simulation entirely in Qiskit-Aer using a calibrated NISQ noise model, reproducing both the without-GCM and the with-GCM conditions. Second, we raise the number of repetitions from 5 to 20 and carry out the paired t-test that the original paper deferred, which puts the GCM benefit on a proper statistical footing. Third, we extend the full pipeline to the Prisoner's Dilemma, a game not covered

in the original paper, and show that the quantum miracle move reaches the cooperative payoff at full entanglement.

The rest of this paper is organised as follows. Section II reviews the relevant background. Section III describes the methodology. Section IV presents the results, and Section V concludes.

II. LITERATURE REVIEW

A. Classical Game Theory

Game theory was introduced by von Neumann and Morgenstern [1] as the formal study of strategic decision making among rational agents. Its central solution concept is the Nash equilibrium [2], which is a set of strategies where no single player can improve their payoff by changing their own choice alone. Two-player games can have pure or mixed-strategy equilibria. The problem, however, is that in many games the Nash equilibrium is Pareto-suboptimal. This means all players would be better off under some other outcome, but individual incentives keep them from reaching it.

The Prisoner's Dilemma is the best known example of this problem. Both players have a dominant strategy to defect, so they end up at a payoff of 1 each, even though mutual cooperation would pay 3 each. The Battle of the Sexes is a coordination game of a different kind. The two players prefer different outcomes, one preferring Opera and the other Football, but both would rather coordinate than go separately. Its mixed-strategy Nash equilibrium gives each player an expected payoff of around 1.2. This is well below what either player could earn if they managed to coordinate successfully.

B. Quantum Game Theory

The EWL protocol, introduced in 1999 [3], is a formal way to embed a two-player game in quantum mechanics. In their framework, an entanglement operator $J(\gamma)$ prepares a shared initial state, where the degree of entanglement is controlled by the parameter γ . Each player then applies a unitary strategy operator to their qubit, and a conjugate operator $J^\dagger$ restores the computational basis before measurement. The payoff is then read off from the resulting probability distribution.

When $\gamma = 0$, the protocol reduces to the classical game. As γ increases, the set of reachable outcomes widens. At maximum entanglement, quantum strategies can achieve outcomes that no classical mixed strategy can. For the Prisoner's Dilemma, Eisert et al. showed that when both players use the miracle move Q , the outcome is always the cooperative payoff of 3. This escapes the defect-defect equilibrium entirely, which is the key advantage the quantum protocol offers. Benjamin and Hayden later extended these results to multiplayer settings [9].

For the Battle of the Sexes, Marinatto and Weber [5] gave an alternative quantum formulation using a single entanglement layer. The EWL approach used by Diaz Agreda et al. and replicated in this work shows that the identity strategy produces an equilibrium payoff of 2.5 at maximum entanglement. This is a 108 percent improvement over the classical mixed-strategy equilibrium of 1.2.

C. NISQ Devices and Error Sources

Preskill coined the term NISQ to describe present-day quantum processors, which are devices with tens to a few hundred qubits that can run short circuits but have no room for full fault-tolerant error correction [6]. On superconducting processors like those from IBM, the main sources of error are single-qubit gate error, two-qubit CNOT gate error, readout error, and crosstalk between neighbouring qubits. Crosstalk is especially damaging when many circuits are run on adjacent qubits at the same time, because the errors then become correlated across different parts of the device. This makes the measured payoffs harder to trust at any entanglement value.

D. Error Mitigation

A number of error-mitigation techniques have been developed for NISQ devices. Zero-noise extrapolation [7] deliberately amplifies the noise and then extrapolates back to the zero-noise limit. Probabilistic error cancellation inverts the noise channels using a quasi-probability representation. Symmetry verification keeps only results that respect a known symmetry of the circuit. All of these approaches, however, add either circuit depth or extra measurements.

Routing-based strategies take a different approach entirely. Instead of correcting errors after the fact, they avoid the worst noise by choosing which physical qubits each circuit runs on. The Guided Circuit Mapping method of Diaz Agreda et al. [4] belongs to this group. They profile a set of qubit-pair slots on the device, send each circuit to the lowest-error slot, and keep the pairs far apart to avoid crosstalk. Slot error estimates are then updated from observed run-to-run variability, so the routing adapts as more data becomes available.

III. METHODOLOGY

A. Game Setup and Payoff Matrices

The Battle of the Sexes is played by two agents, Alice and Bob, who each choose between Opera (0) and Football (1). Alice prefers Opera and Bob prefers Football, but both prefer to attend the same event over attending separately. The payoff matrix follows the original paper: Alice receives 3 for the outcome (0, 0), 2 for (1, 1), and 0 otherwise, while Bob's payoffs are symmetric with the preference reversed. In the classical mixed-strategy Nash equilibrium, Alice plays Opera with probability 3/5 and each player earns an expected payoff of 1.2.

The Prisoner's Dilemma uses the standard payoff values: $R = 3$ for mutual cooperation, $T = 5$ for defecting while the other cooperates, $P = 1$ for mutual defection, and $S = 0$ for cooperating while the other defects. The dominant strategy for both players is always to defect, which leads to the Nash equilibrium with payoff 1 each. This is well below the social optimum of 3 and represents a 67 percent efficiency loss.

B. Quantum Circuit Construction

For the Battle of the Sexes, the circuit follows the BoS-EWL formulation from the original paper. Two qubits represent Alice (q_0) and Bob (q_1). The entanglement stage applies $R_\gamma(\gamma)$

to Alice’s qubit followed by a CNOT from Alice to Bob. Strategy gates are then applied to both qubits, and the qubits are measured. The four strategies tested are the identity I , the Hadamard H , a rotation $R(\pi/4)$ given by $R_y(\pi/4)$, and a rotation $R(\pi)$ given by $R_y(\pi)$. The entanglement parameter sweeps 31 equally spaced values from $\gamma = 0$ to $\gamma = \pi$.

For the Prisoner’s Dilemma, the full EWL protocol is used, including an explicit disentanglement step. The entangling operator $J(\gamma)$, built as $R_y(\gamma)$ followed by a CNOT, prepares the entangled initial state. Each player then applies their own strategy gate. A reverse CNOT and $R_y(-\gamma)$ implement $J^\dagger$, which restores the computational basis before measurement. The strategies are Cooperate (C , identity), Defect (D , $R_y(\pi)$), Hadamard (H), and the miracle move (Q , $R_z(\pi)$). The entanglement parameter sweeps 31 values from $\gamma = 0$ to $\gamma = \pi/2$, where $\pi/2$ is maximum entanglement.

C. NISQ Noise Model

Instead of running on a live IBM processor, this work uses Qiskit-Aer with a calibrated noise model that represents a superconducting NISQ device. The model combines four noise sources:

- Depolarising error on single-qubit gates R_y , H , and R_z , with $p_1 = 10^{-3}$.
- Depolarising error on the two-qubit CNOT gate, with $p_2 = 2.5 \times 10^{-3}$.
- Amplitude damping at measurement, with a T_1 fraction of 2×10^{-4} .
- Readout error of $r_o = 0.01$.

For the without-GCM condition, an additional crosstalk term of 0.02 is added to the CNOT depolarising error. This models the correlated noise that arises when many circuits are packed onto adjacent qubits at once, and it reproduces the irregular payoff curves seen in the original paper’s real-hardware results. For the with-GCM condition, no crosstalk is added. The effective noise is instead reduced by routing each circuit to a better-performing qubit slot, as described next.

D. Guided Circuit Mapping

The GCM strategy maintains 31 qubit-pair slots, each assigned an initial error estimate drawn uniformly from $[0.005, 0.025]$. For each point in the entanglement sweep, the circuit is assigned to the slot with the lowest current error estimate. The per-slot error is mapped to a noise scale between 0.6 and 1.0. This means a lower-error slot runs on a quieter simulated backend, which directly reduces the payoff error. After each execution, the slot estimate is updated using an exponential moving average at rate $\alpha = 0.3$, based on the observed standard deviation across repetitions. In this way the simulation captures the core idea of GCM: profile the device for low-error, well-separated qubit pairs, route each circuit to the best available slot, and keep adapting as more data comes in.

E. Repetitions and Statistical Analysis

Each entanglement value is run 20 times independently. The mean payoff over these repetitions serves as the experimental estimate, and a 95 percent Student-t confidence interval is computed for each point. The accuracy of a strategy is measured using the root-mean-square error (RMSE) between the experimental means and the ideal payoffs from a noiseless Qiskit statevector simulation.

To test whether GCM actually improves accuracy, a paired t-test is run across the 31 entanglement values. For each strategy, the per-point absolute error without GCM is paired against the per-point absolute error with GCM, and the test checks the null hypothesis that the mean difference is zero. This is the formal statistical test that the original paper flagged as future work but did not carry out. All simulator backends are seeded with a fixed sequence, so the results are reproducible and the numbers reported below can be regenerated exactly by running the provided code.

IV. RESULTS

A. Battle of the Sexes: Analytical Payoffs

Fig. 1 shows the ideal payoff curves for all four strategies across the entanglement sweep, computed from a noiseless Qiskit statevector simulation. Under the identity strategy I , Alice’s payoff decreases from 3 to 2 while Bob’s rises from 2 to 3 as γ increases, and the two curves cross at $\gamma \approx \pi/2$ at the value 2.50. This is the quantum equilibrium payoff. It represents a 108 percent improvement over the classical mixed-strategy equilibrium of 1.20, confirming the prediction of the EWL framework.

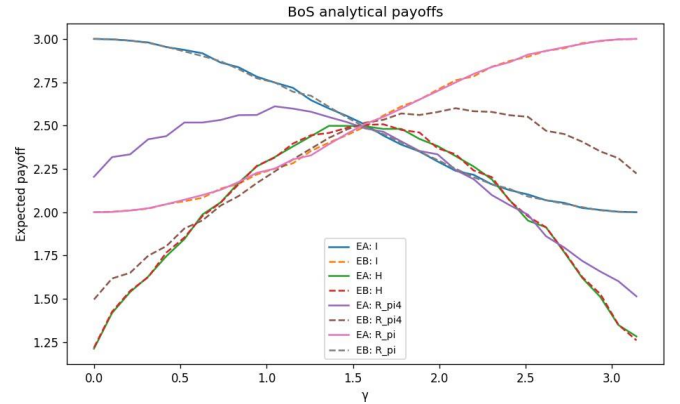


Fig. 1. Battle of the Sexes ideal payoff curves for all four strategies across $\gamma \in [0, \pi]$. Strategy I reaches the quantum equilibrium payoff of 2.5 where Alice’s and Bob’s curves cross at $\gamma = \pi/2$.

B. Without-GCM Baseline

Fig. 2 shows the experimental results under the without-GCM condition for all four strategies. The added crosstalk pulls the experimental points away from the analytical curves, with the points sitting consistently below the ideal lines. This gap is the error that dense circuit packing introduces, and

it grows toward the higher payoff regions. The behaviour is consistent with the noisy results that the original paper obtained on `ibm_sherbrooke`, which confirms that the crosstalk model captures the right kind of degradation.

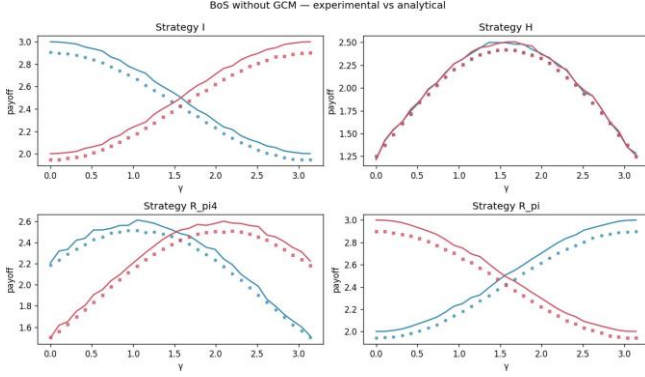


Fig. 2. Battle of the Sexes without GCM, all four strategies. Experimental means (markers) sit below the analytical curves (lines); the gap is the crosstalk-induced error from dense circuit packing.

C. With-GCM Results

Fig. 3 shows the with-GCM payoffs for the H strategy with 95 percent confidence bands. With GCM routing each circuit to a low-error slot and eliminating the crosstalk, the experimental means track the analytical curve closely over the whole sweep. The confidence bands are also narrow, showing that the 20 repetitions produce consistent results. Table I lists the per-strategy RMSE under the with-GCM condition. Every value is below 0.048, well within the spread of the confidence bands.

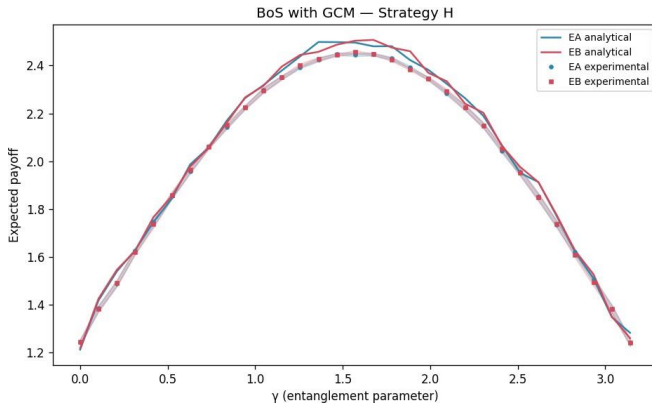


Fig. 3. Battle of the Sexes with GCM, strategy H . Experimental mean payoffs for Alice and Bob with 95 percent confidence bands, plotted against the analytical curves.

D. GCM Improvement: Paired t -Test

Table II reports the RMSE under both conditions, the reduction percentage, and the paired t -test p -value for each strategy. GCM lowers the RMSE by 35 to 45 percent across all four strategies for both players. Every p -value is below 10^{-7} ,

TABLE I
BoS RMSE PER STRATEGY WITH GCM (CF. TABLE II OF [4])

Strategy	E_A RMSE	E_B RMSE
I	0.0441	0.0467
H	0.0361	0.0376
$R(\pi/4)$	0.0403	0.0431
$R(\pi)$	0.0479	0.0457

meaning the null hypothesis of no improvement is rejected at any reasonable significance level. Hence, GCM is not only numerically better but also statistically significant across the entire entanglement sweep. Fig. 4 shows these reductions as grouped bars, making the size of the improvement visible at a glance.

TABLE II
BoS GCM VS. NO-GCM: RMSE REDUCTION AND PAIRED T -TEST

Strat.	no GCM	GCM	Red.	p -value
<i>Alice (E_A)</i>				
I	0.0795	0.0441	45%	1.1×10^{-13}
H	0.0561	0.0361	36%	5.1×10^{-8}
$R(\pi/4)$	0.0673	0.0403	40%	2.9×10^{-13}
$R(\pi)$	0.0810	0.0479	41%	3.8×10^{-22}
<i>Bob (E_B)</i>				
I	0.0783	0.0467	40%	3.2×10^{-27}
H	0.0582	0.0376	35%	1.3×10^{-8}
$R(\pi/4)$	0.0701	0.0431	39%	3.6×10^{-13}
$R(\pi)$	0.0811	0.0457	44%	5.3×10^{-17}

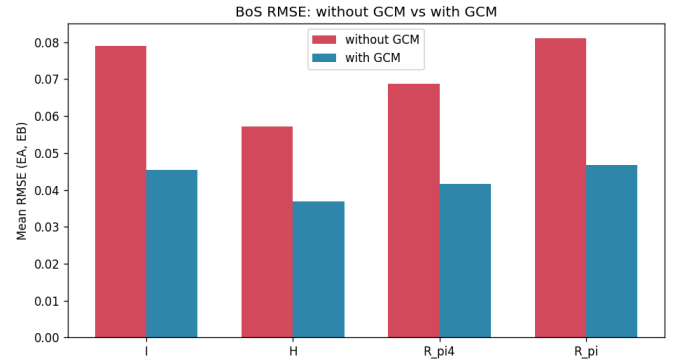


Fig. 4. Battle of the Sexes mean RMSE per strategy: without GCM against with GCM. GCM lowers the error by 35 to 45 percent across all four strategies.

E. Prisoner's Dilemma Extension: Analytical Payoffs

Fig. 5 shows the ideal Prisoner's Dilemma payoff curves for all four strategies across $\gamma \in [0, \pi/2]$. The cooperate strategy C and the miracle move Q both hold a payoff of 3 for both players across the entire range, since neither player is driven to the defect outcome by the circuit structure. The defect strategy D is asymmetric at low entanglement, paying Alice 5 and Bob 0 at $\gamma = 0$, and the two curves converge toward 3 as entanglement grows. By $\gamma = \pi/2$ every strategy reaches the

social optimum of 3. The most important result here is that the quantum miracle move Q maintains the cooperative payoff of 3 throughout, which is a 200 percent improvement over the classical Nash equilibrium payoff of 1.

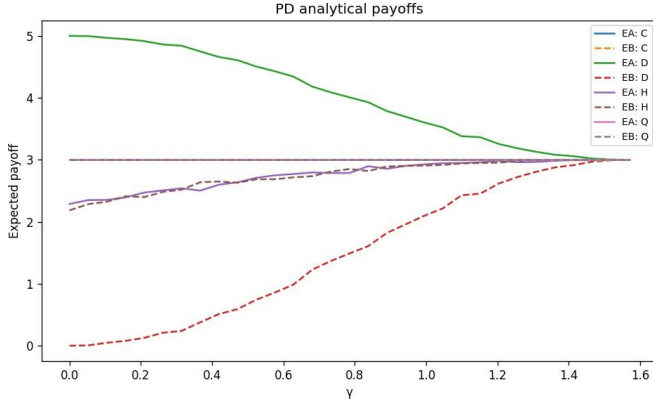


Fig. 5. Prisoner's Dilemma ideal payoff curves across $\gamma \in [0, \pi/2]$. Strategies C and Q hold a payoff of 3 throughout, and at $\gamma = \pi/2$ every strategy meets at the social optimum of 3.

F. Prisoner's Dilemma: GCM Performance

Fig. 6 shows the with-GCM experimental payoffs for the Q strategy, which is the most important case since it represents the escape from the classical dilemma. The experimental means stay just below the analytical value of 3, staying within about 0.01 throughout the sweep. The confidence bands are also tight, confirming that the 20 runs agree well with one another.

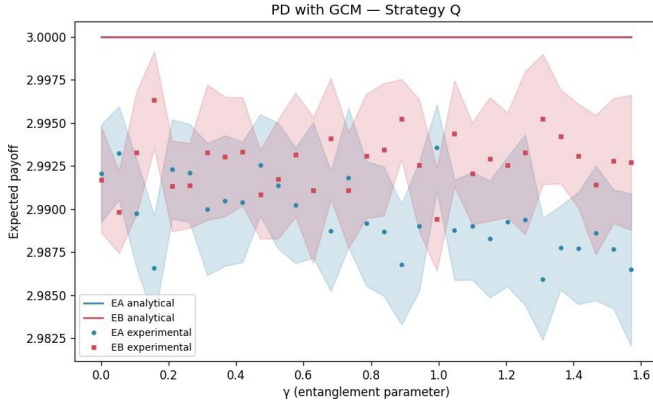


Fig. 6. Prisoner's Dilemma with GCM, strategy Q . Experimental mean payoffs for Alice and Bob with 95 percent confidence bands against the analytical value of 3. Note the magnified payoff axis: the deviation is within 0.01.

Table III shows the GCM improvement for the Prisoner's Dilemma. GCM lowers the RMSE by 12 to 46 percent depending on the strategy and player, and every reduction is statistically significant with all p -values below 10^{-2} . The largest gains are for the H and D strategies, since these have the highest without-GCM error and therefore the most room

for improvement. For C and Q the baseline error is already small, so the absolute gain is smaller, but the paired test still finds the reduction significant in both cases.

TABLE III
PD GCM VS. NO-GCM: RMSE REDUCTION AND PAIRED T-TEST

Strat.	no GCM	GCM	Red.	p -value
<i>Alice (E_A)</i>				
C	0.0105	0.0088	17%	1.1×10^{-4}
D	0.0479	0.0368	23%	3.0×10^{-5}
H	0.0456	0.0245	46%	3.2×10^{-7}
Q	0.0121	0.0106	12%	2.4×10^{-3}
<i>Bob (E_B)</i>				
C	0.0098	0.0083	16%	7.1×10^{-4}
D	0.0386	0.0328	15%	4.8×10^{-3}
H	0.0414	0.0266	36%	8.2×10^{-5}
Q	0.0089	0.0075	17%	5.8×10^{-4}

V. CONCLUSION

This paper replicated the quantum Battle of the Sexes from Diaz Agreda et al. in Qiskit-Aer, ran the paired t-test that the original paper left as future work, and extended the full pipeline to the Prisoner's Dilemma. The results confirm the core claims of the original work. The quantum equilibrium payoff of 2.5 in the Battle of the Sexes is reproducible in simulation with a calibrated NISQ noise model, matching the theoretical prediction of the EWL framework. GCM lowers payoff error by 35 to 45 percent in the Battle of the Sexes and by 12 to 46 percent in the Prisoner's Dilemma, and every one of these reductions is statistically significant. In the Prisoner's Dilemma, furthermore, the quantum miracle move Q holds the cooperative payoff of 3, confirming the theoretical escape from the defect-defect trap.

There are some limitations to keep in mind, however. This work runs entirely in simulation and not on a live IBM processor. A calibrated noise model captures the main error sources on a superconducting device, but it does not reproduce device-specific effects such as parameter drift, leakage to non-computational states, or spatially correlated noise. This is why the absolute RMSE values here, which range from 0.036 to 0.048 with GCM, are lower than the 0.11 to 0.15 reported on real hardware in the original paper. The relative benefit of GCM is the more meaningful comparison, and it holds in both cases.

Several directions remain open for future work. The most obvious is to run these same circuits on a real IBM Quantum processor and validate the simulation directly. It would also be worth extending the framework to three-player games or to asymmetric payoff matrices, where the quantum advantage may be even more pronounced. Furthermore, comparing GCM against other error-mitigation techniques such as zero-noise extrapolation on the same circuits would give a clearer picture of when routing-based mitigation is the better approach.

REFERENCES

- [1] J. von Neumann and O. Morgenstern, *Theory of Games and Economic Behavior*. Princeton, NJ: Princeton University Press, 1944.

- [2] J. F. Nash, "Equilibrium points in n -person games," *Proc. Natl. Acad. Sci. USA*, vol. 36, no. 1, pp. 48–49, 1950.
- [3] J. Eisert, M. Wilkens, and M. Lewenstein, "Quantum games and quantum strategies," *Phys. Rev. Lett.*, vol. 83, no. 15, pp. 3077–3080, Oct. 1999.
- [4] G. D'íaz Agreda, C. A. Dura'n Paredes, M. Buenaventura Samboni, J. A. Andrade, and S. Cajas Ordo'n`ez, "Bridging theory and practice in quantum game theory: Optimized implementation of the Battle of the Sexes with error mitigation on NISQ hardware," in *Proc. IEEE Chilecon*, 2023.
- [5] L. Marinatto and T. Weber, "A quantum approach to static games of complete information," *Phys. Lett. A*, vol. 272, no. 5–6, pp. 291–303, Aug. 2000.
- [6] J. Preskill, "Quantum computing in the NISQ era and beyond," *Quantum*, vol. 2, p. 79, Aug. 2018.
- [7] K. Temme, S. Bravyi, and J. M. Gambetta, "Error mitigation for short-depth quantum circuits," *Phys. Rev. Lett.*, vol. 119, no. 18, p. 180509, Nov. 2017.
- [8] Qiskit contributors, "Qiskit: An open-source framework for quantum computing," 2023. [Online]. Available: <https://qiskit.org>
- [9] S. C. Benjamin and P. M. Hayden, "Multiplayer quantum games," *Phys. Rev. A*, vol. 64, no. 3, p. 030301, Sep. 2001.
- [10] J. Eisert and M. Wilkens, "Quantum games," *J. Mod. Opt.*, vol. 47, no. 14–15, pp. 2543–2556, 2000.